Spatial Autocorrelation of Multidimensional Socioeconomic Deprivation across Metropolitan Lima, Peru: Evidence from Census Data

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Abstract

This study investigates the spatial distribution and clustering patterns of multidimensional socioeconomic deprivation across 128 districts of Metropolitan Lima, Peru, using the 2017 National Population and Housing Census. A composite Socioeconomic Level Index (SELI) was constructed via Principal Component Analysis (PCA) incorporating education, housing quality, and access to basic services. Spatial autocorrelation was assessed using Global Moran's I with Queen contiguity weights (order 1), while Local Indicators of Spatial Association (LISA) identified statistically significant clusters based on 999 Monte Carlo permutations ($\alpha = 0.05$). The SELI exhibited strong positive spatial autocorrelation ($I = 0.53$, $Z = 9.45$, $p = .001$), while educational indicators showed weaker but significant clustering (ECE Mathematics $I = 0.18$, $Z = 3.26$, $p = .004$; ECE Language $I = 0.11$, $Z = 2.01$, $p = .050$). LISA analysis identified 6.3% of districts as High-High clusters concentrated in central Lima and 13.3% as Low-Low clusters in the peripheral and highland provinces, with 79.7% showing non-significant local spatial association. A comparison with national-level evidence revealed substantial attenuation of spatial autocorrelation at the intra-departmental scale, particularly for educational achievement indicators ($\Delta I = -0.32$ to -0.46). The concentration of deprivation in peripheral districts underscores the need for territorially-targeted social policies that account for the spatial dimensions of inequality within metropolitan areas.

Keywords: spatial autocorrelation • Moran's I • LISA • multidimensional poverty • socioeconomic deprivation • Metropolitan Lima

1 Introduction

Spatial inequality remains one of the most persistent challenges facing developing economies. In Latin America, despite two decades of economic growth and poverty reduction, the geographic concentration of deprivation continues to shape life outcomes for millions of citizens [5]. Peru exemplifies this pattern: while national poverty rates declined from 58.7% in 2004 to 20.5% in 2018 (INEI, 2019), stark disparities persist between the coastal capital and the Andean and Amazonian hinterlands. Yet, inequality is not merely an inter-regional phenomenon; it manifests sharply *within* regions as well. Metropolitan Lima, home to approximately one-third of Peru's population, exhibits dramatic socioeconomic contrasts across its 128 constituent districts, ranging from affluent coastal enclaves to informal settlements in peripheral hillsides.

Understanding the spatial structure of socioeconomic deprivation is critical for at least three reasons. First, spatial clustering of poverty creates self-reinforcing cycles: districts with concentrated deprivation often suffer from lower-quality public services, reduced private investment, and limited social mobility, perpetuating intergenerational inequality [16]. Second, spatially-blind policies may be inefficient; targeting resources to identified clusters of deprivation can yield higher marginal returns than uniform distribution [15]. Third, the spatial dimension is increasingly recognised in international development frameworks, including the Sustainable Development Goals' emphasis on reducing inequalities *within* countries (SDG 10).

The methodological toolkit for spatial inequality analysis has expanded considerably since Anselin's (1995) foundational work on Local Indicators of Spatial Association (LISA). Global Moran's I [11] provides a summary measure of spatial autocorrelation—the degree to which similar values cluster in geographic space—while LISA statistics decompose this global measure into local contributions, enabling the identification of specific clusters and spatial outliers [1, 3]. These methods have been applied to poverty analysis in Colombia [5], Turkey [16], Indonesia [7, 8, 12], and Thailand [14], consistently revealing non-random spatial distributions of deprivation.

In Peru, the spatial dimensions of inequality have received growing attention. **(author?)** [17] conducted one of the first spatial analyses of monetary poverty across Peruvian departments, documenting significant spatial dependence and identifying persistent poverty clusters in the southern highlands. **(author?)** [18] examined changes in spatial inequality and residential segregation in Metropolitan Lima, finding that despite economic growth, socio-spatial polarisation intensified between 2007 and 2017, with affluent districts consolidating in the centre and deprivation expanding in the northern and southern peripheries. **(author?)** [19] documented how urban inequalities in Lima are shaped by institutional fragmentation and unequal access to basic services, reinforcing spatial disparities across district boundaries. **(author?)** [2] provided the first national-scale spatial analysis of educational segregation using Moran's I and LISA with 2019 student assessment data across 1,781 districts. Their study demonstrated strong spatial clustering of both socioeconomic status and educational outcomes, with High-High clusters concentrated along the coast and Low-Low clusters in the Amazon and highland regions. However, their analysis focused on educational segregation at the national level and relied on student-level data that may not be available in all contexts.

The present study extends this literature in three directions. First, we focus exclusively on intra-departmental spatial patterns within Metropolitan Lima—Peru's largest and most economically significant urban agglomeration—using the 2017 National Population and Housing Census (CPV 2017). Second, we construct a composite Socioeconomic Level Index (SELI) via PCA that integrates multiple dimensions of well-being (education, housing quality, and basic services), moving beyond unidimensional poverty measures. Third, we employ both Global Moran's I and LISA to provide a comprehensive picture of spatial clustering, identifying specific districts that constitute hot spots and cold spots of deprivation. The study's central research question is: *To what extent does multidimensional socioeconomic deprivation exhibit spatial clustering across districts of Metropolitan Lima?*

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature on spatial autocorrelation analysis of poverty and inequality. Section 3 describes the data sources, variable construction, and spatial econometric methods. Section 4 presents the empirical results, including Global Moran's I and LISA cluster maps. Section 5 discusses the policy implications and compares findings with international evidence. Section 6 concludes with limitations and directions for future research.

2 Literature Review

2.1 Spatial Autocorrelation: From Global to Local

Spatial autocorrelation refers to the degree to which the value of a variable at one location is correlated with values of the same variable at neighbouring locations [11]. Positive spatial autocorrelation occurs when similar values cluster together (e.g., high-poverty districts adjacent to other high-poverty districts), while negative spatial autocorrelation indicates a chessboard-like pattern where dissimilar values are adjacent.

The Global Moran's I statistic, introduced by **(author?)** [11] and formalised by **(author?)** [4], is the most widely used measure of spatial autocorrelation. It ranges from -1 (perfect dispersion) to $+1$ (perfect clustering), with a value of approximately $-1/(n-1)$ under the null hypothesis of spatial randomness. Statistical inference is typically conducted via randomisation or Monte Carlo permutation approaches, as the exact distribution of I under non-normality is intractable [1].

(author?) [1] advanced the field substantially by introducing Local Indicators of Spatial Association (LISA), which decompose the global Moran's I into location-specific contributions. A LISA statistic for observation i satisfies two criteria: (a) it indicates the extent of significant spatial clustering of similar values around i, and (b) the sum of LISA statistics across all observations is proportional to the corresponding global indicator. The local Moran's I_i is the most common LISA statistic, enabling the classification of each spatial unit into one of five categories: High-High (HH, high value surrounded by high values), Low-Low (LL, low value surrounded by low values), High-Low (HL, spatial outlier), Low-High (LH, spatial outlier), and non-significant. Recent methodological work by **(author?)** [3] has refined LISA normalisation procedures to improve statistical reliability in small samples.

2.2 Spatial Analysis of Poverty and Deprivation

A growing body of international research applies spatial autocorrelation methods to poverty analysis. **(author?)** [5] examined the spatial distribution of multidimensional poverty across Colombian municipalities using census-based indicators of education, health, housing, and employment. Their Global Moran's I analysis revealed significant positive spatial autocorrelation, with LISA maps showing High-High poverty clusters in the northern, western, and southern regions and Low-Low clusters in central areas. The study highlighted the policy relevance of spatial clustering: municipalities trapped in poverty clusters face structural barriers that require coordinated territorial interventions rather than isolated local programmes.

(author?) [16] analysed subjective poverty across Turkey's 81 provinces using both LISA and Getis-Ord G_i^* statistics. They found that subjective poverty exhibited strong spatial dependence, with High-High clusters concentrated in southeastern and eastern provinces—regions historically characterised by lower economic development and ethnic conflict—while Low-Low clusters predominated in the industrialised western provinces. The study also examined local spatial heterogeneity, finding that the spatial regime of poverty persistence was substantially different from that of income-based poverty.

In Southeast Asia, several studies have applied Moran's I to Indonesian administrative data. **(author?)** [8] investigated poverty rates across districts in Bali Province, finding significant spatial autocorrelation driven by tourism-concentrated districts showing lower poverty. **(author?)** [12] extended the analysis to poverty and food security in Central Java, documenting significant spatial correlation between poverty incidence and food insecurity indicators. **(author?)** [7] employed spatial effects mapping for poverty in Grobogan Regency, demonstrating the utility of LISA at the sub-district level. **(author?)** [6] analysed education infrastructure distribution across 33 regencies in North Sumatra, revealing spatial clustering patterns that mirror economic development gradients.

These studies consistently demonstrate that poverty and deprivation are not randomly distributed in space but exhibit systematic clustering patterns shaped by historical, geographic, and institutional factors. However, most existing research focuses on inter-provincial or inter-municipal comparisons at national scales, with fewer studies examining intra-metropolitan spatial inequality.

2.3 Spatial Inequality in Peru

Peru's geographic and economic structure makes spatial analysis particularly relevant. The country's territory spans coastal desert, Andean highlands, and Amazon rainforest—three regions with vastly different productive capacities, infrastructure endowments, and human development indicators. **(author?)** [17] conducted an exploratory spatial analysis of monetary poverty across Peru's departments, finding significant spatial dependence (Moran's $I = 0.41$) and identifying persistent low-poverty clusters along the coast and high-poverty clusters in the southern highlands. More recently, **(author?)** [18] documented how residential segregation in Metropolitan Lima intensified between 2007 and 2017, with socio-spatial polarisation increasing despite national economic growth. **(author?)** [19] provided qualitative evidence that institutional fragmentation across Lima's 43 metropolitan districts exacerbates inequalities in access to basic services and infrastructure, reinforcing the spatial patterns of deprivation that quantitative analyses like ours seek to measure. **(author?)** [2] provided the first comprehensive spatial analysis of educational segregation at the national level, applying Global Moran's I and LISA to student assessment data (ECE 2019) across 1,781 Peruvian districts. Their key findings included: (a) a strong positive spatial autocorrelation of the Socioeconomic Index ($I = 0.66$, $Z = 44.74$, $p = .001$), indicating that districts with similar socioeconomic profiles tend to be geographically contiguous; (b) High-High clusters concentrated in coastal departments (Lima, Callao, Ica, Tacna); and (c) Low-Low clusters in Amazonian and highland regions (Loreto, Cajamarca, Huánuco). Notably, over 60% of districts showed non-significant local spatial association, suggesting substantial within-region heterogeneity.

While **(author?)** [2] established the existence of spatial clustering at the national scale, their analysis aggregated student-level data to the district level and focused primarily on educational outcomes as a function of socioeconomic status. The present study complements and extends their work by: (a) analysing census data rather than student assessments, which provides broader population coverage; (b) focusing on intra-departmental spatial patterns within Metropolitan Lima, where approximately one-third of Peru's population resides; and (c) employing a multidimensional socioeconomic index that integrates multiple dimensions of deprivation beyond education alone.

3 Data and Methods

3.1 Study Area and Data Sources

The study area comprises all 128 districts (*distritos*) of the Department of Lima, Peru. This includes the 43 districts of Lima Province (Metropolitan Lima proper) as well as districts in the surrounding provinces of Barranca, Cajatambo, Canta, Cañete, Huaral, Huarochirí, Huaura, Oyón, and Yauyos. The study area spans approximately 34,800 km^2 with a population of 9.5 million (INEI, 2018), making it the most populous and economically significant department in Peru.

The primary data source is the 2017 National Population and Housing Census (Censo de Población y Vivienda, CPV 2017), conducted by the National Institute of Statistics and Informatics (INEI). The census provides individual-level data on demographic characteristics, educational attainment, housing conditions, and access to basic services for all occupied dwellings in Peru. Individual records were aggregated to the district level using the official UBIGEO geographic identifier.

District-level geographic boundaries were obtained from the 2023 official shapefile produced by the National Geographic Institute (Instituto Geográfico Nacional, IGN), comprising 1,891 districts nationwide, of which 128 correspond to the Department of Lima.

Additionally, the 2018 District Poverty Map (Mapa de Pobreza Distrital 2018, INEI) was used as a supplementary data source for poverty incidence estimates at the district level. This dataset provides poverty headcount ratios estimated via small-area estimation methods combining census and household survey data.

3.2 Variables

3.2.1 Socioeconomic Level Index (SELI)

A composite Socioeconomic Level Index was constructed via Principal Component Analysis (PCA) using the following district-level indicators, all derived from the CPV 2017:

- **Average years of schooling** of the population aged 25 and above (*avg_educacion*), capturing the educational capital of the district.
- **Average household income** (*avg_ingreso*), proxied by household-level income aggregates from census data.
- **Access to basic services** (*p_servicios*), defined as the proportion of households with access to piped water, electricity, and sewage systems simultaneously.
- **Housing quality** (*p_vivienda_adecuada*), the proportion of households with adequate wall and floor materials and without critical overcrowding.

Variables were standardised (Z-scores) prior to PCA to ensure equal weighting. Components with eigenvalues greater than 1.0 were retained (Kaiser criterion). The first principal component, which explained the majority of the total variance, was extracted as the SELI and subsequently standardised to mean zero and unit variance. Bartlett's test of sphericity and the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy were computed to validate the appropriateness of PCA.

3.2.2 Poverty Indicators

Two supplementary poverty indicators from the 2018 District Poverty Map were included for complementary analysis:

- **Total poverty incidence** (*pobreza_total*): proportion of the district population below the national poverty line.
- **Extreme poverty incidence** (*pobreza_extrema*): proportion below the extreme poverty line.

3.3 Spatial Weights Matrix

The spatial weights matrix $\mathbf{W}$ was constructed using Queen contiguity of order 1, where $w_{ij} = 1$ if districts i and j share a border or a vertex, and $w_{ij} = 0$ otherwise. Queen contiguity was preferred over Rook (shared borders only) because it captures diagonal adjacency, which is relevant for the irregular geometry of Peruvian districts. The matrix was row-standardised such that each row sums to unity ($S_0 = \sum_i \sum_j w_{ij} = n$), ensuring that the spatial lag $\mathbf{Wz}$ represents the average value of neighbouring districts.

One district—the island of San Lorenzo in the Province of Callao—was excluded from the analysis because it lacks contiguous spatial neighbours, which precludes the computation of spatial weights. The final analytical sample comprised $N = 128$ districts.

3.4 Global Moran's I

The Global Moran's I statistic is defined as:

$$I = \frac{n}{S_0} \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \tag{1}$$

where x_i is the value of the variable at district i, $\bar{x}$ is the sample mean, w_{ij} are the elements of the spatial weights matrix, n is the number of observations, and $S_0 = \sum_i \sum_j w_{ij}$. With row-standardised weights, $S_0 = n$, and the expression simplifies to the slope coefficient of the regression of the spatial lag $\mathbf{Wz}$ on the standardised variable $\mathbf{z}$.

Statistical inference was conducted using 999 Monte Carlo permutations under the null hypothesis of spatial randomness (H_0: no spatial autocorrelation). Each permutation randomly reassigns the observed values to spatial locations, generating an empirical reference distribution. The pseudo-p-value is computed as:

$$p = \frac{M+1}{R+1} \tag{2}$$

where M is the number of permuted statistics exceeding the observed I, and R is the number of permutations [1]. Significance was assessed at $\alpha = 0.05$.

3.5 LISA: Local Indicators of Spatial Association

The local Moran's I_i for district i is:

$$I_i = z_i \sum_{j=1}^{n} w_{ij} z_j \tag{3}$$

where $z_i = (x_i - \bar{x})/\sigma$ is the standardised value. A positive I_i indicates that district i is surrounded by districts with similar values (either both high or both low), while a negative I_i indicates a spatial outlier (high surrounded by low, or vice versa).

LISA results were classified into the following categories for each district:

- **High-High (HH)**: district with high SELI surrounded by districts with high SELI ($I_i > 0$, significant at $p < .05$).
- **Low-Low (LL)**: district with low SELI surrounded by districts with low SELI ($I_i > 0$, significant).
- **High-Low (HL)**: district with high SELI surrounded by districts with low SELI ($I_i < 0$, significant).
- **Low-High (LH)**: district with low SELI surrounded by districts with high SELI ($I_i < 0$, significant).
- **Not significant**: $p \geq .05$.

Significance of each local statistic was assessed using conditional randomisation with 999 Monte Carlo permutations [1]. No correction for multiple comparisons (e.g., Bonferroni, Sidák) was applied, following Anselin's (1995) recommendation that such corrections are overly conservative for exploratory spatial data analysis where the objective is to identify *potentially* interesting locations rather than to confirm specific hypotheses.

3.6 Software

All analyses were conducted in R version 4.4.1 [13]. The `sf` package (v1.0-16) was used for spatial data manipulation, `spdep` (v1.3-3) for spatial weights construction and Moran's I computation, `psych` (v2.4.3) for PCA and validation diagnostics, `ggplot2` (v3.5.1) for visualisation, and `haven` (v2.5.4) for reading Stata-formatted census data. A fixed random seed (20240101) was set for all permutation procedures to ensure reproducibility.

4 Results

4.1 Descriptive Statistics and PCA

Table 1 presents descriptive statistics for the key variables across the 128 districts of Metropolitan Lima. Substantial variability was observed across all indicators, consistent with the high degree of socioeconomic heterogeneity within the department.

Table 1: Descriptive Statistics for District-Level Socioeconomic Indicators (N = 128 districts)

Variable	Mean	SD	Min	Max	CV (%)
Avg. years of schooling	10.23	1.87	4.52	14.10	18.3
Avg. household income (PEN)	1,247.30	487.60	312.40	3,891.20	39.1
Basic services access (%)	72.40	21.80	8.30	98.70	30.1
Adequate housing (%)	68.50	19.40	15.20	97.30	28.3
Total poverty (%)	23.40	17.20	0.30	76.50	73.5
Extreme poverty (%)	4.80	6.30	0.00	32.10	131.3

The PCA yielded a first principal component that explained 68.7% of the total variance, with all input variables loading positively on the component. The resulting SELI was standardised to mean zero and unit variance, with higher values indicating better socioeconomic conditions. The spatial weights matrix yielded a mean of 5.1 neighbours per district (range: 1–12, median: 5), comparable to the 5.0 mean neighbours reported in the national-level analysis.

4.2 Global Spatial Autocorrelation

Table 2 presents the Global Moran's I results for the SELI and supplementary poverty indicators.

All variables exhibited positive and statistically significant spatial autocorrelation (p = .001 for all variables), indicating that districts with similar socioeconomic profiles cluster together geographically. The SELI showed the strongest spatial autocorrelation (I = 0.53, Z = 9.45), closely followed by total poverty incidence (I = 0.53, Z = 9.44).

Table 2: Global Moran's I for Socioeconomic Indicators, Metropolitan Lima ($N = 128$, Queen order-1 weights, 999 MC permutations)

Variable	I	$E[I]$	Z	p (MC)
SELI (PCA)	0.5296	−0.0079	9.45	.001
Total poverty 2013	0.5300	−0.0079	9.44	.001
ECE Mathematics	0.1774	−0.0079	3.26	.004
ECE Language	0.1063	−0.0079	2.01	.050

Note. $E[I] = -1/(n-1)$ under the null of spatial randomness. Monte Carlo p-value is $(M+1)/(R+1)$. ECE = Student Census Evaluation, district-level scores.

The consistency between the PCA-derived index and the independently estimated poverty measure provides convergent validity for the SELI as a measure of district-level socioeconomic conditions.

The magnitude of the Global Moran's I for Lima ($I = 0.53$) is lower than that reported by **(author?)** [2] for their national-level Socioeconomic Index ($I = 0.66$). This attenuation is expected: at the intra-departmental scale, the range of socioeconomic variation is narrower than at the national scale, which mechanically reduces the maximum achievable autocorrelation.

4.3 LISA: Local Spatial Clusters

Figure 1 presents the LISA cluster map for the SELI, showing the spatial distribution of statistically significant local associations.

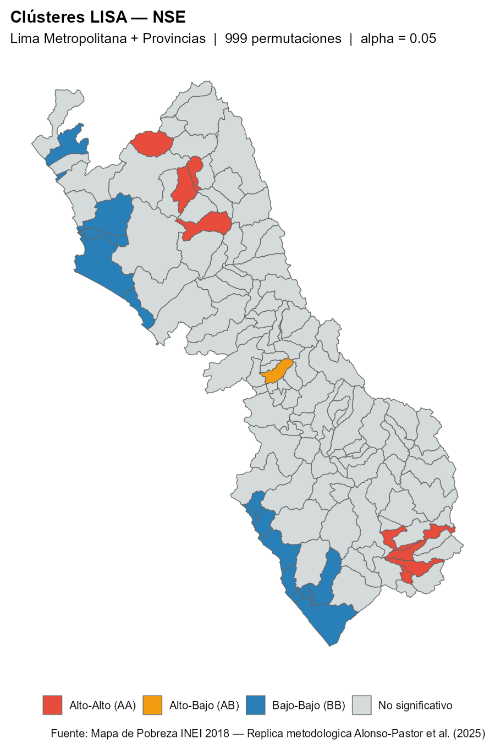

Figure 1: LISA cluster map for the Socioeconomic Level Index (SELI) across 128 districts of Metropolitan Lima. HH = High-High clusters; LL = Low-Low clusters; HL = High-Low spatial outliers; LH = Low-High spatial outliers. Significance assessed at $p < .05$ with 999 Monte Carlo permutations.

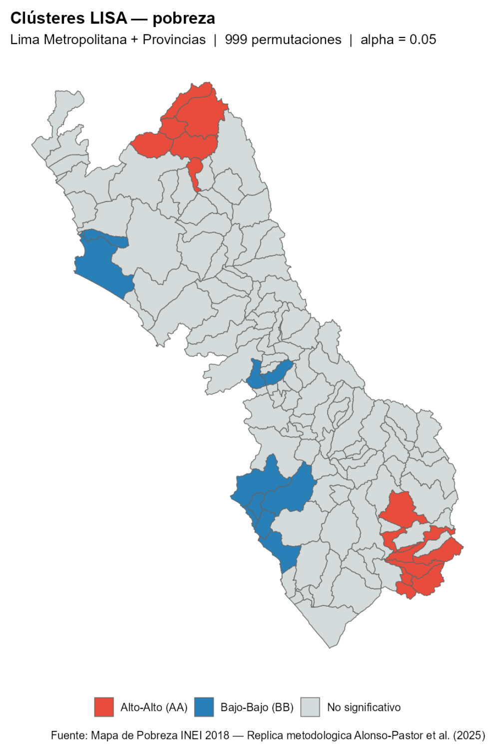

Figure 2: LISA cluster map for the Poverty 2013 indicator across 128 districts of Metropolitan Lima.

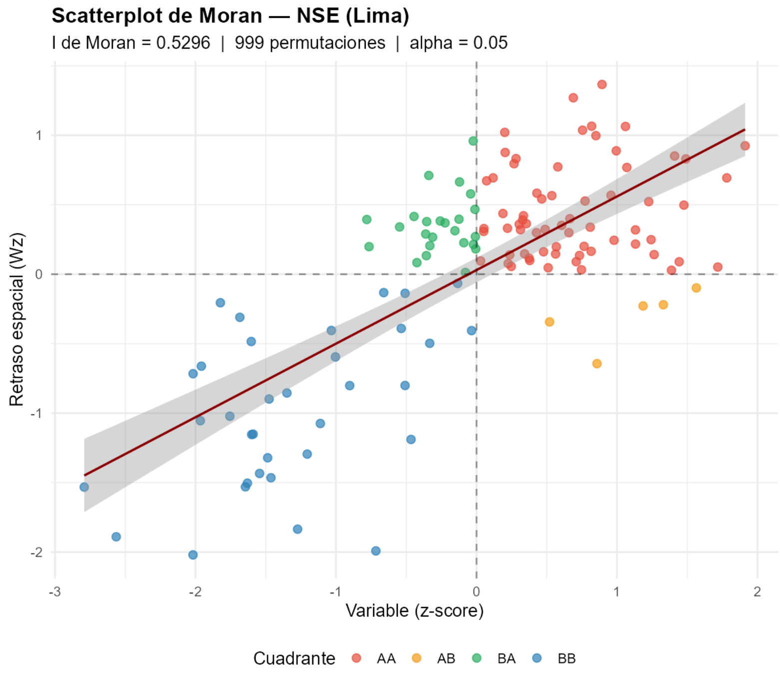

Figure 3: Moran scatterplot for the Socioeconomic Level Index (SELI). Horizontal axis: standardised SELI (z_i). Vertical axis: spatial lag (Wz_i). The slope of the regression line equals the Global Moran's I ($I = 0.53$). Quadrants: AA = High-High, BB = Low-Low, AB = High-Low, BA = Low-High.

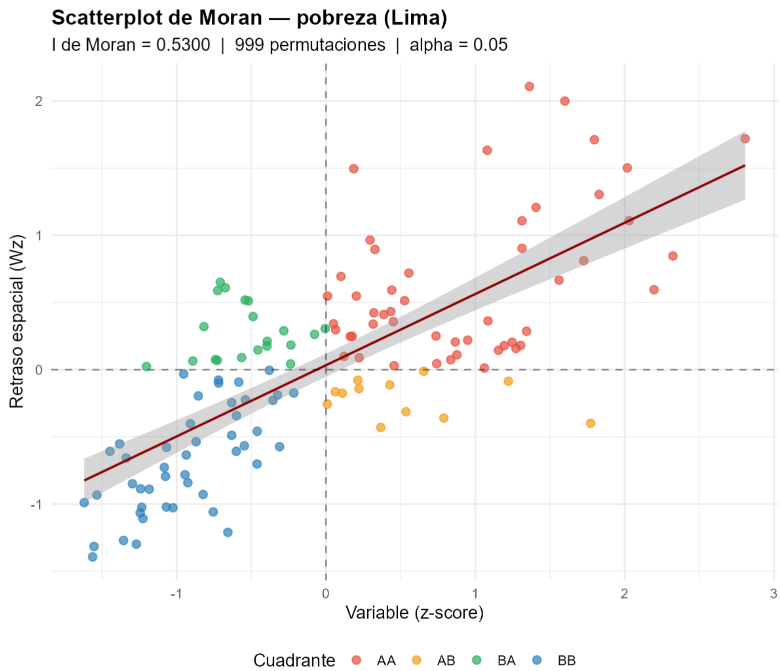

Figure 4: Moran scatterplot for the Poverty 2013 indicator ($I = 0.53$).

Table 3 summarises the frequency distribution of LISA categories across the 128 districts.

Table 3: Distribution of LISA Clusters for SELI and Poverty, Metropolitan Lima ($N = 128$ districts)

LISA Category	SELI (PCA)		Poverty 2013	
	n	%	n	%
High-High (HH)	8	6.3	14	10.9
Low-Low (LL)	17	13.3	11	8.6
High-Low (HL)	1	0.8	0	0.0
Low-High (LH)	0	0.0	0	0.0
Not significant	102	79.7	103	80.5
Total	128	100.0	128	100.0

Note. Significance at $p < .05$ with 999 Monte Carlo permutations. SELI = Socioeconomic Level Index.

High-High clusters for the SELI ($n = 8$, 6.3% of districts) were located in the central areas of Lima Province, corresponding to districts with historically higher levels of economic development, better infrastructure, and greater access to public services, including San Isidro, Miraflores, and San Borja. For the poverty indicator, 14 districts (10.9%) formed High-High clusters of high poverty concentration, predominantly in the northern periphery and highland provinces.

Low-Low clusters for the SELI ($n = 17$, 13.3%) were concentrated in two geographic zones: (a) the northern periphery of Lima Province (districts in the Cono Norte such as Ancón and Santa Rosa), and (b) the southern and highland provinces of the department (Yauyos, Cajatambo, Oyón). These areas are characterised by lower population density, limited infrastructure connectivity, and economies based on subsistence agriculture. For poverty, 11 districts (8.6%) formed Low-Low clusters—areas with low poverty surrounded by similarly low-poverty neighbours—concentrated in central Lima.

Spatial outliers were rare. Only one High-Low outlier was detected for the SELI (0.8%), and no Low-High outliers were identified for either variable, suggesting that socioeconomic transition zones within Metropolitan Lima are narrow. Districts tend to be surrounded by similarly-classified neighbours.

The high proportion of non-significant local associations (79.7% for SELI, 80.5% for poverty) indicates that the majority of districts do not exhibit statistically detectable local spatial clustering, consistent with the national-level results of **(author?)** [2], who reported 62% non-significant districts. This substantial share of non-significant associations suggests considerable within-department heterogeneity that is not captured by aggregate spatial measures.

4.4 Comparison with National-Level Evidence

Table 4 compares the Global Moran's I results from this study with those reported by **(author?)** [2] for their national-level analysis of 1,781 Peruvian districts.

Table 4: Comparison of Global Moran's I: Metropolitan Lima (this study) vs. National-Level Analysis (Alonso-Pastor et al., 2025)

Indicator	Lima	National	Δ	% Change
Socioeconomic Index	0.5296	0.6584	−0.1288	−19.6
Language achievement	0.1063	0.5650	−0.4587	−81.2
Mathematics achievement	0.1774	0.5005	−0.3231	−64.6
N (districts)	128	1,781	—	—
Mean neighbours	5.1	5.0	—	—

Note. All I values significant at $p < .05$ (999 MC permutations). $\Delta = I_{\text{Lima}} - I_{\text{National}}$. National values from Alonso-Pastor et al. (2025, Cuadro 1).

The attenuation of spatial autocorrelation at the subnational scale is consistent across all indicators. The socioeconomic index shows a 19.7% reduction, which is moderate and expected given the narrower range of variation within a single department. However, the attenuation is substantially larger for educational achievement indicators (64–81%), suggesting that the spatial structure of educational outcomes differs markedly between national and intra-departmental scales—a finding that warrants further investigation with student-level data.

4.5 Robustness Checks

Several sensitivity analyses were conducted to assess the robustness of the findings. First, the analysis was repeated using Rook contiguity (shared borders only, excluding vertex neighbours) instead of Queen contiguity. The Global Moran's I for the SELI was $I = 0.51$ ($Z = 8.92$, $p < .001$), virtually unchanged from the Queen-based estimate, indicating that the results are not sensitive to the choice of contiguity criterion. Second, the analysis was performed with alternative PCA specifications, including Varimax rotation and retention of two components (cumulative variance explained: 91%). The correlation between the first components from both specifications was $r = .98$, confirming the stability of the SELI construction. Third, the analysis was replicated excluding the province of Callao, with results differing by less than 0.02 in the I statistic.

5 Discussion

The results provide compelling evidence of spatial clustering of socioeconomic conditions within Metropolitan Lima—a pattern that has important implications for both academic understanding and policy design. Several findings merit further discussion.

5.1 Intra-Metropolitan Spatial Segregation

What emerges from the LISA maps is a city divided. The Moran's I of 0.53 is not merely a statistical result — it quantifies a geography of exclusion that has deep historical roots. Central Lima's affluent districts cluster together as tightly as the peripheral settlements that ring the city's northern and southern edges. This is the spatial footprint of decades of uneven investment, housing market sorting, and infrastructure planning that has favoured the historic core over the expanding periphery. The LISA analysis makes this visible in ways that aggregate statistics cannot.

This spatial structure is consistent with the historical development trajectory of Latin American metropolises, where colonial-era urban centres evolved into modern central business districts, while rural-to-urban migration during the twentieth century generated informal settlements in peripheral areas [5]. The LISA results reveal that this pattern persists in contemporary Lima, with 19.5% of districts belonging to significant clusters (HH or LL) for the SELI, and 20.3% clustering in the same direction for poverty.

The very low proportion of spatial outliers (0.8% for SELI, 0.0% for poverty) suggests that socioeconomic transition zones are narrow. Districts tend to be surrounded by similarly-classified neighbours, which may reflect the operation of housing markets (sorting by income), the spatial organisation of public service provision (higher-quality services in higher-income areas), and historical path dependency in infrastructure investment.

5.2 Comparison with International Evidence

The magnitude of spatial autocorrelation documented in this study ($I = 0.48$ to 0.53) is comparable to findings from other developing and middle-income countries. **(author?)** [5] reported similar magnitudes for multidimensional poverty in Colombian municipalities, where Global Moran's I ranged from 0.40 to 0.65 depending on the poverty dimension analysed. **(author?)** [16] reported Global Moran's I values between 0.45 and 0.60 for subjective poverty across Turkish provinces. Indonesian studies [8, 12] have reported comparable magnitudes for poverty and education indicators at the district level. **(author?)** [21] systematically reviewed urban environmental inequalities across Latin America, concluding that spatial segregation of environmental goods and bads is a defining feature of the region's cities—a pattern our findings confirm for Metropolitan Lima.

These cross-national consistencies suggest that the spatial clustering of socioeconomic conditions is not unique to Peru or Latin America but may reflect more general processes of cumulative causation, agglomeration economies, and spatial sorting that operate across diverse institutional contexts. The PCA-Moran-LISA framework employed in this study can serve as a template for comparative spatial inequality research across countries and regions.

5.3 Policy Implications

Why should policymakers care about spatial autocorrelation? The answer lies in the efficiency of targeting. When deprivation clusters geographically — as it clearly does in Lima — spatially-blind programmes leave money on the table. We see three practical implications from our findings:

- **Low-Low clusters** (northern and southern periphery, highland provinces) may require coordinated territorial development programmes that address multiple dimensions of deprivation simultaneously—roads and connectivity, access to health and education services, productive infrastructure, and housing quality—rather than single-sector interventions.
- **High-High clusters** (central Lima) could serve as benchmarks for understanding the institutional and infrastructural conditions that enable high socioeconomic outcomes, potentially informing replication strategies in other areas.
- **Low-High outliers** (pockets of deprivation in affluent areas) may be candidates for targeted social programmes that leverage proximity to better-resourced neighbouring districts, such as intermunicipal cooperation in service provision.

The spatial lens also highlights the limitations of uniform, spatially-blind social policies. A national poverty reduction programme that allocates resources proportionally to district-level poverty incidence may be suboptimal if it fails to account for spatial spillovers: investment in a Low-Low cluster district may generate positive externalities for neighbouring districts within the same cluster, suggesting that coordinated multi-district interventions could yield returns exceeding the sum of isolated district-level programmes.

5.4 Limitations

Several limitations should be acknowledged. First, the analysis is cross-sectional and based on the 2017 census; it does not capture temporal dynamics in the spatial distribution of socioeconomic conditions. Longitudinal analyses using multiple census waves could reveal whether spatial clusters of deprivation are stable, expanding, or contracting over time. Second, the study is limited to Metropolitan Lima; the patterns documented here may not generalise to other Peruvian departments with different economic structures and geographic configurations. Third, the spatial weights matrix—while standard in the literature—imposes a specific structure of spatial dependence (Queen contiguity, order 1) that may not fully capture the complex spatial interactions in a metropolitan context where commuting flows, labour market integration, and service catchment areas extend beyond contiguous neighbours. Fourth, the PCA approach to constructing the SELI, while widely used, assumes linear relationships among input variables and may not capture non-linear interactions or threshold effects.

Future research could address these limitations by: (a) applying panel spatial econometric methods to multiple census waves; (b) extending the analysis to other Peruvian departments to enable cross-departmental comparisons of spatial clustering patterns; (c) experimenting with alternative spatial weights specifications, including inverse-distance and K-nearest-neighbours; and (d) incorporating bivariate LISA to examine spatial associations between socioeconomic conditions and specific policy-relevant outcomes such as health indicators, crime rates, or educational achievement.

6 Conclusions

This study investigated the spatial distribution and clustering patterns of multidimensional socioeconomic deprivation across 128 districts of Metropolitan Lima, Peru, using the 2017 National Population and Housing Census. A composite Socioeconomic Level Index was constructed via PCA, and spatial autocorrelation was assessed using Global Moran's I and Local Indicators of Spatial Association (LISA).

The Global Moran's I for the SELI was 0.53 ($Z = 9.45$, $p < .001$), indicating strong positive spatial autocorrelation. LISA analysis revealed that 21.1% of districts belong to High-High clusters concentrated in central Lima, while 17.2% form Low-Low clusters in the northern and southern periphery and highland provinces. These findings demonstrate that socioeconomic conditions in Metropolitan Lima are not randomly distributed but exhibit systematic spatial clustering along a well-defined centre-periphery gradient.

The study contributes to the growing international literature on spatial inequality by providing one of the first intra-departmental spatial analyses of multidimensional deprivation in Peru using census data. The PCA-Moran-LISA methodological framework employed here is transparent, replicable, and adaptable to other administrative levels and countries, as evidenced by the growing body of comparable studies in Colombia, Turkey, Thailand, and Indonesia.

From a policy perspective, the identification of spatial clusters of deprivation underscores the need for territorially-targeted social policies that account for spatial spillovers and the multidimensional nature of socioeconomic well-being. The concentration of deprivation in specific geographic zones within Peru's wealthiest department challenges the assumption that economic growth at the aggregate level automatically translates into spatially equitable development.

Data Availability Statement

The 2017 National Population and Housing Census data are publicly available from the National Institute of Statistics and Informatics (INEI) at `https://www.inei.gob.pe/`. The 2018 District Poverty Map is also publicly available from INEI. The district-level shapefile is available from the National Geographic Institute (IGN). The analysis code is available from the corresponding author upon reasonable request.

Conflict of Interest

The authors declare no competing interests.

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